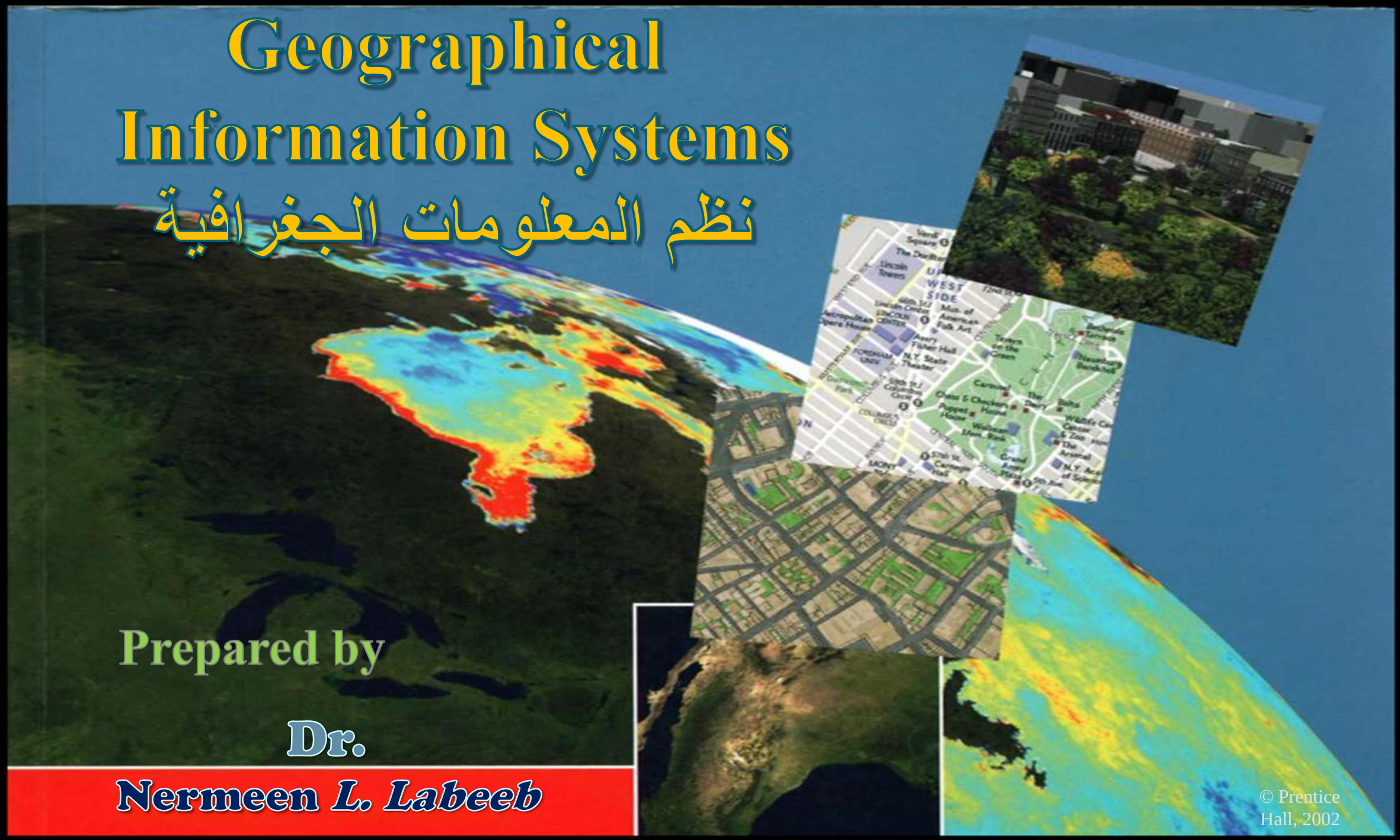
Geographical Information Systems

نظم المعلومات الجغرافية



Prepared by

Dr.

Nermeen L. Labeeb

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PART 1 Fundamentals of GIS

CHAPTER 1

What is GIS?

By the end of this chapter you should be able to:

- Explain what **GIS** is.
- Outline the **components and elements of GIS**.
- Give examples of the **applications** of **GIS**.
- Describe how the **real world** is represented in **GIS**.
- Provide examples of the **analysis GIS** can perform.
- Geospatial data representation models

1.5 GEOSPATIAL DATA

- By definition, ***geospatial data*** cover the ***location*** of ***spatial features***.
- To locate ***spatial features*** on the ***Earth's surface***, we can use either a ***geographic*** or a ***projected coordinate system***.

1. **<u>A geographic coordinate system</u>**: is expressed in ***longitude*** and ***latitude.***

(Figure 1)
Geographic Coordinate System

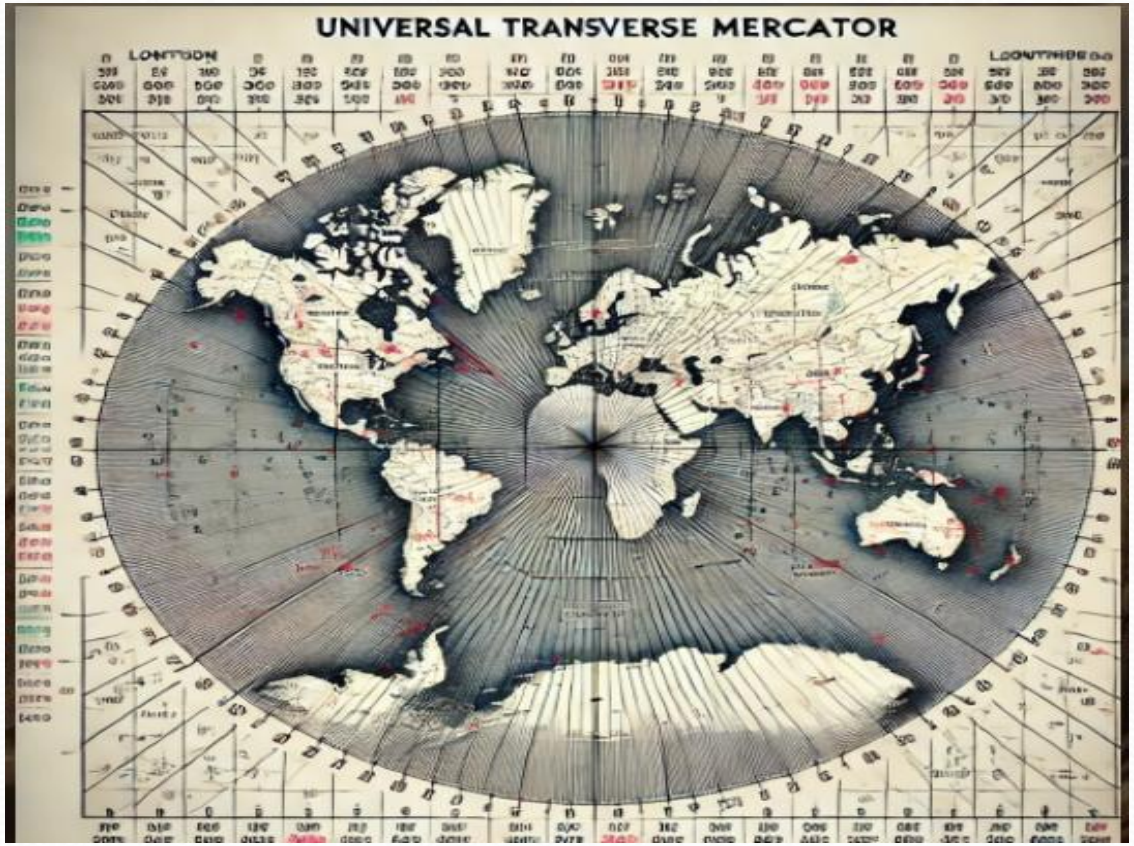
(Figure 2)
Projected Coordinate System

2. <u>A projected coordinate system</u>: is expressed in ***x, y coordinates***.

- Many ***projected coordinate systems*** are available for use in ***GIS***.
- An example is the ***Universal Transverse Mercator*** (***UTM***) ***grid system***, which divides the Earth's surface between **84°N** and **80°S** into **60 zones**.

- A basic principle in GIS is that ***map layers*** representing different ***geospatial data*** must align spatially;
  - in other words, they are based on the same ***spatial reference***.

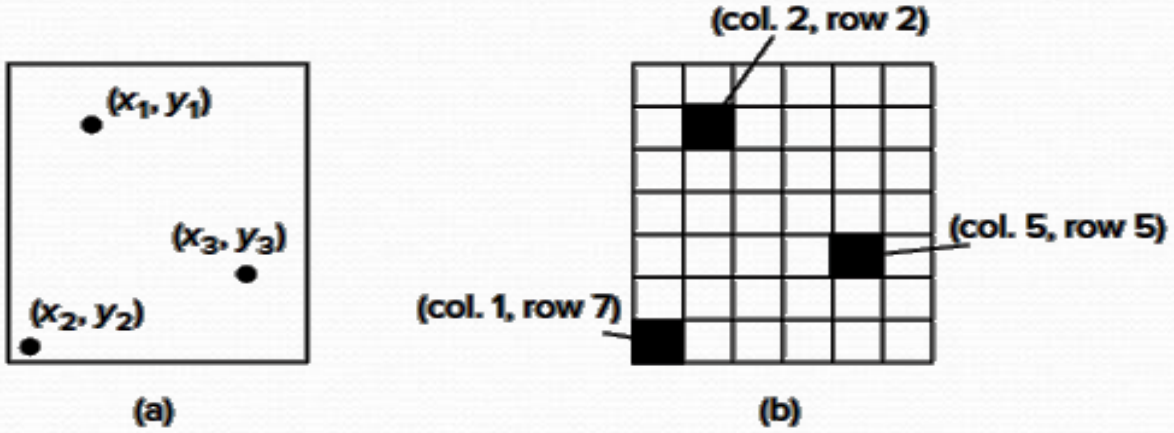
- A GIS represents ***geospatial data*** as either: (*Figure 4*)
  - ***Vector data*** or
  - ***Raster data***



(Figure 3)
Universal Transverse Mercator (UTM) grid system

1. **The vector data model**: uses *points*, *lines*, and *polygons* to represent *spatial features* with a clear *spatial location* and *boundary* such as **streams**, **land parcels**, and **vegetation stands** (*Figure 5*).
   - Each *feature* is assigned an **ID** *so that it can be associated with its attributes*.

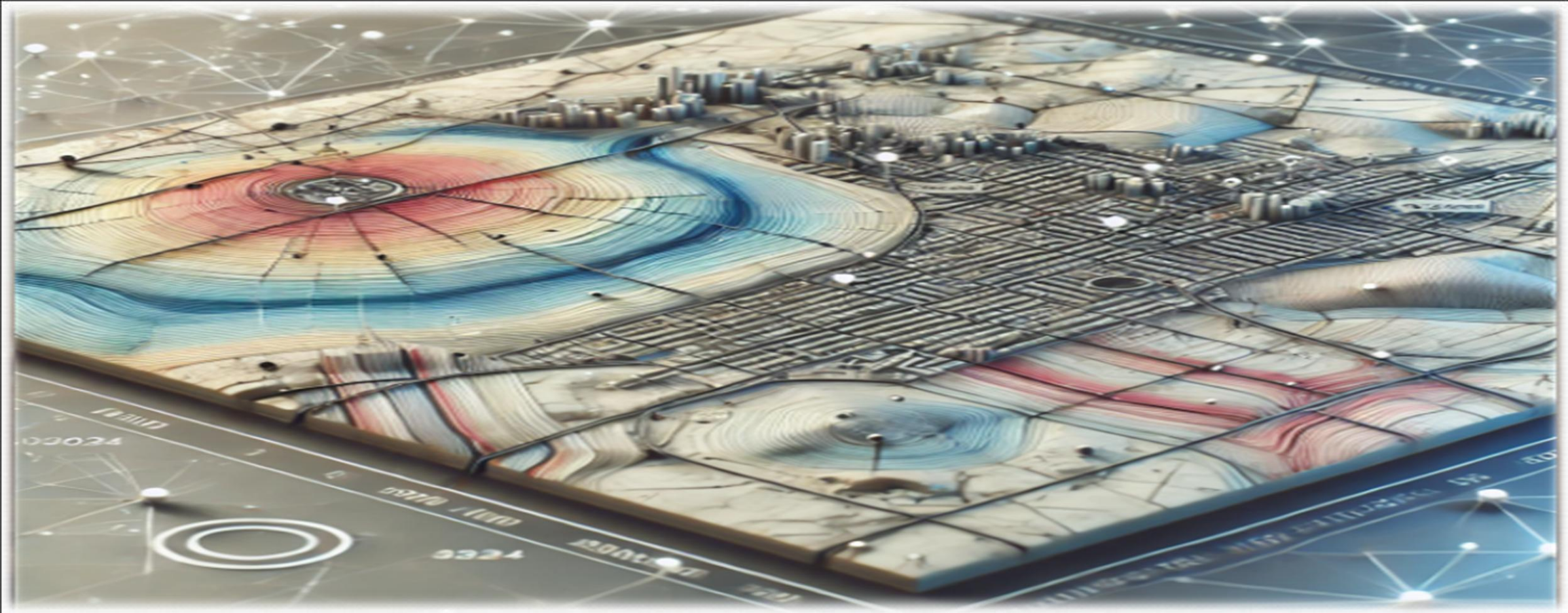
2. **The raster data model**: uses a *grid* and *grid cells* to represent *spatial features*:
   - **point features** are represented by *single cells*,
   - **line features** by *sequences of neighboring cells*, and
   - **polygon features** by *collections of contiguous cells*.



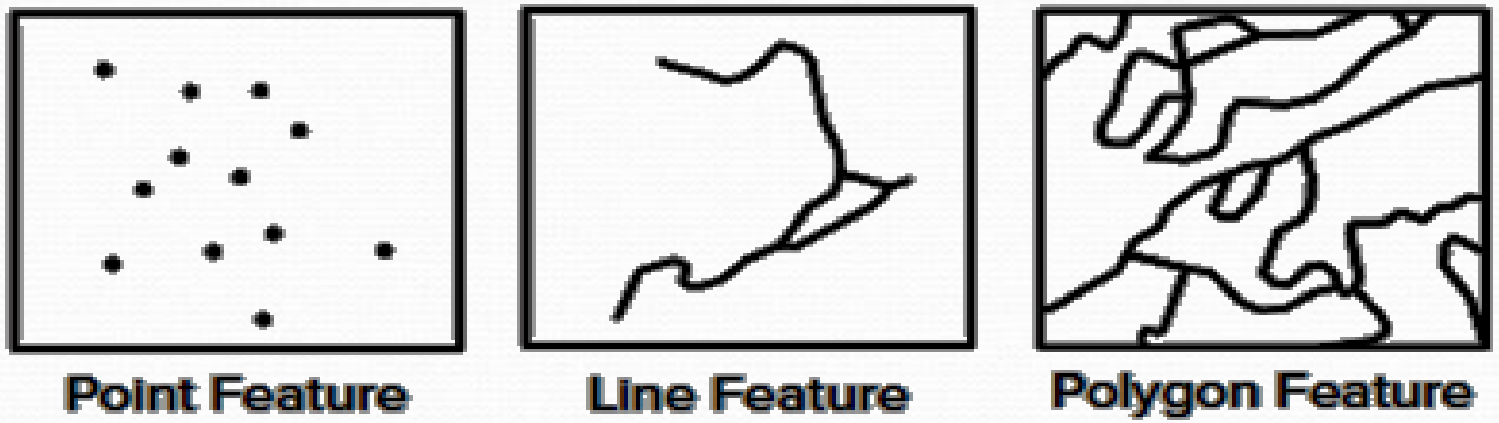
(Figure 4)

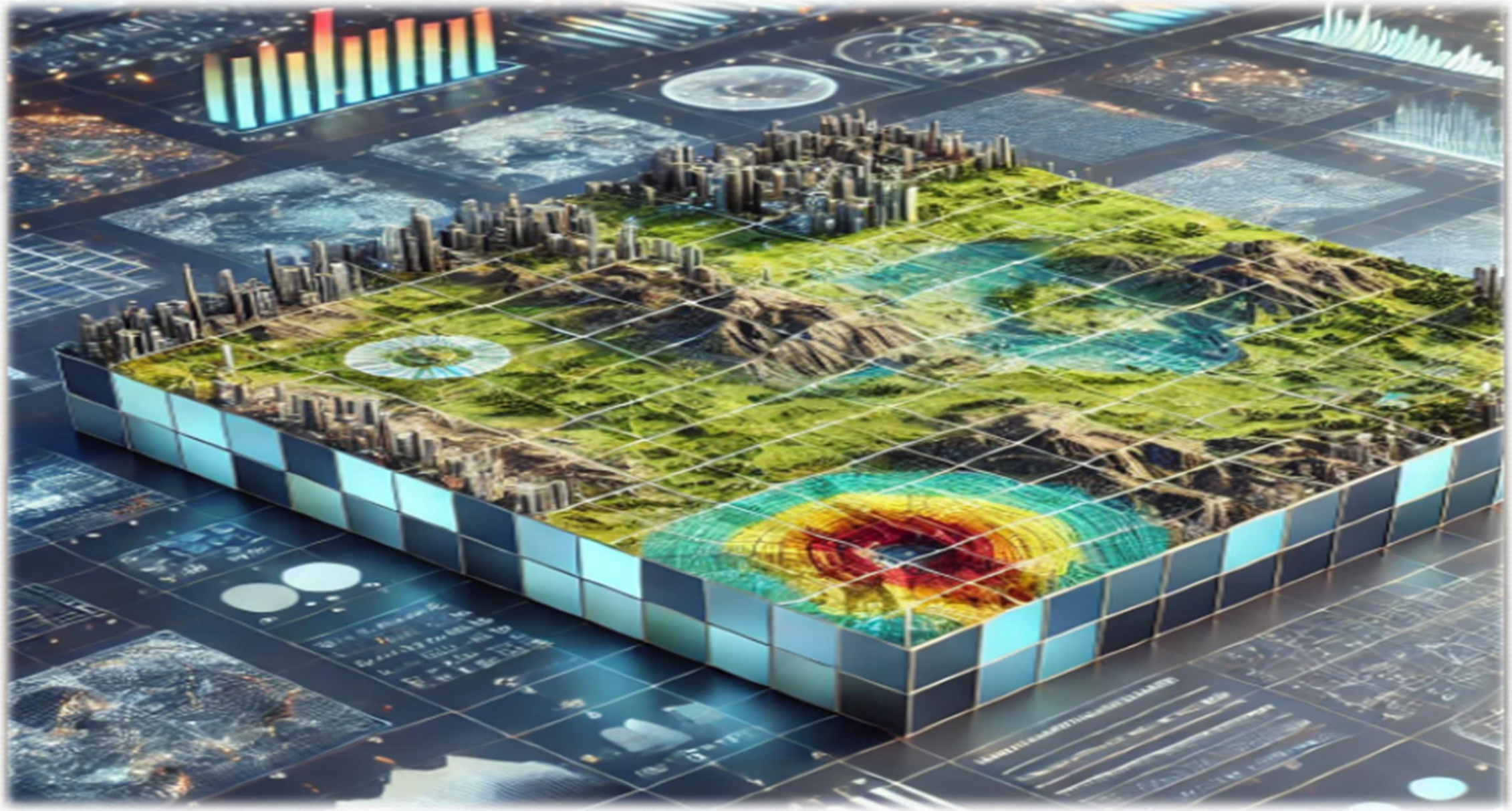
(a) Vector Data Model uses x, y coordinates to represent point feature.
(b) The raster data model uses cells in a grid to represent point feature.

- The *cell value* corresponds to the *attribute* of the *spatial feature* at the *cell location*.
- *Raster data* are ideal for *continuous features* such as *elevation* and *temperature* (*Figure 7*).



(Figure 5)
Point, line, and polygon features

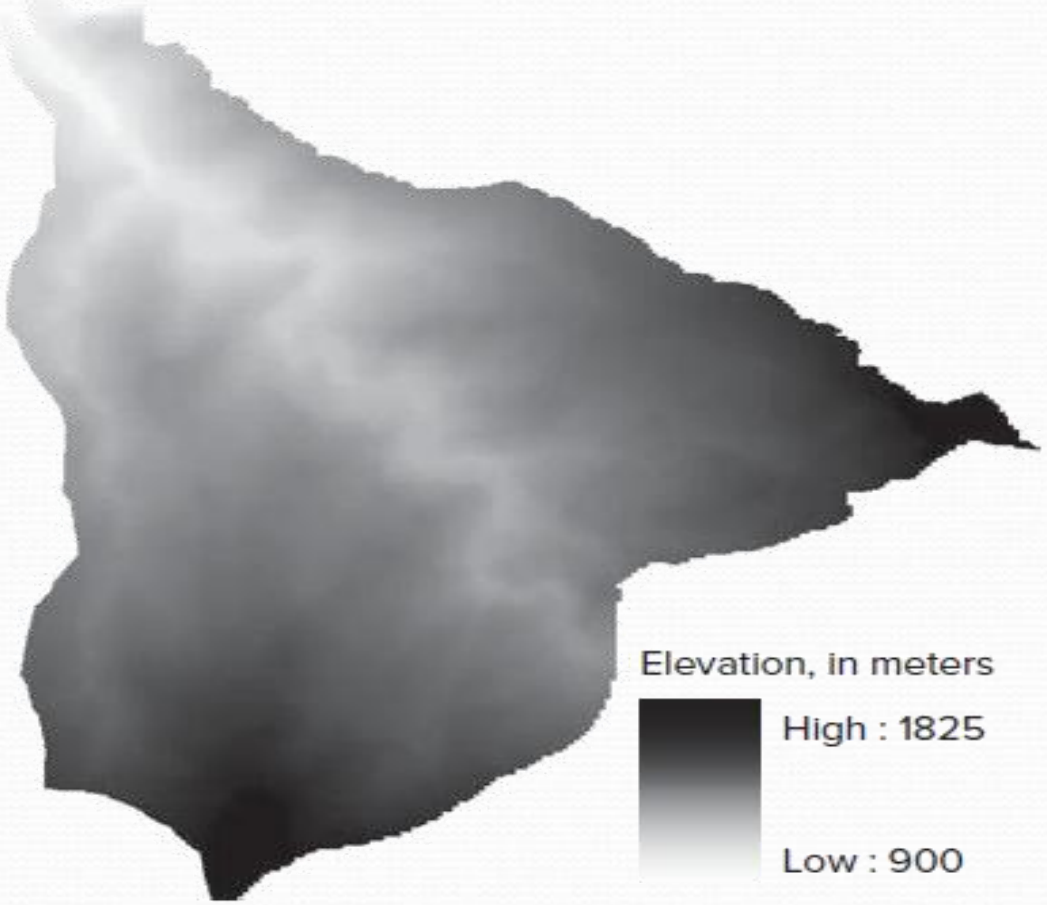




(Figure 6)
Raster Data Model

- A ***vector data model*** can be ***georelational*** or ***object-based.***
- The ***georelational vector data model*** is used in (GIS) to represent ***spatial features*** and their relationships with ***geographic locations***.
- In this model we use:

1. **<u>Geometry</u>:** It defines the **shape** and **location** of ***spatial features***. ***Points***, ***lines***, and ***polygons*** are common geometric primitives used to represent ***features*** like ***cities***, ***roads***, or ***land parcels***.
2. **Attributes**: the ***attribute data*** associated with the ***spatial features***.



(Figure 7)
A Raster-based Elevation Layer

- These ***attributes*** provide *<u>additional information about the features</u>*, such as **population** of a city, **length** of a road, or **land use type** in a parcel.

3. **Coordinate System**: to reference the ***geographic locations*** of the ***features***.
   - This allows for accurate positioning of ***spatial elements*** on the Earth's surface.
4. **Topology** : Topology explicitly expresses the ***spatial relationships*** and ***connectivity*** between ***features***.
   - This helps in maintaining the ***integrity*** of ***spatial data***, ensuring ***consistency*** in representations like **adjacent polygons** sharing boundaries.
5. **Relational Database Structure**: The ***georelational model*** is implemented using ***Relational Database Management Systems (RDBMS).***
   - ***Tables*** are used to store both ***geometric*** and ***attribute data***.
   - ***Relationships*** between different ***tables*** are established to maintain the link between ***spatial*** and ***attribute information***.

- This model provides a **structured** and **efficient** way to **organize**, **manage**, and **analyze** ***spatial data***, making it a foundational component in GIS applications.

End of Lecture_03